Advanced Material Physics

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Part I

Dielectric Polarization under Static Electric Field

Unit system used in this note. Throughout this note, the unit system is chosen such that $4\pi\epsilon_0 = 1$.

1 Multipole Expansion of a Point Cloud

Potential of a point cloud and the need for an expansion. Consider N point charges q_i at positions $\vec{r}_i$, which together form a point cloud. The electric potential at $\vec{R}$ is the sum of the contributions of all point charges,

$$\phi(\vec{R}) = \sum_{i=1}^{N} \frac{q_i}{|\vec{R} - \vec{r}_i|}.$$

In general, this potential depends on the charge positions in a complicated way. Far from the cloud, however, the multipole expansion breaks the potential down into multipole terms. For brevity, we write $R \equiv |\vec{R}|$ and $r_i \equiv |\vec{r}_i|$ from now on, and $\hat{R} = \vec{R}/R$ denotes the unit vector along $\vec{R}$.

Expansion of the inverse distance in powers of r_i/R. Factoring R out of the distance gives

$$\frac{1}{|\vec{R} - \vec{r}_i|} = \frac{1}{R}\left(1 - 2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}\right)^{-1/2} = \frac{1}{R}\,(1+\eta)^{-1/2}, \qquad \eta = -2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}.$$

Far from the cloud, $r_i \ll R$ for every charge, so $|\eta| \ll 1$ and the Taylor expansion of $(1+\eta)^{-1/2}$ applies:

$$\begin{aligned}(1+\eta)^{-1/2} &= 1 - \frac{1}{2}\eta + \frac{3}{8}\eta^2 + \mathcal{O}(\eta^3) \\ &= 1 - \frac{1}{2}\left(-2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}\right) + \frac{3}{8}\left[4\,\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - 4\,\frac{(\hat{R}\cdot\vec{r}_i)\,r_i^2}{R^3} + \frac{r_i^4}{R^4}\right] + \mathcal{O}(\eta^3).\end{aligned}$$

Grouping the terms by order of r_i/R and dropping the third and higher orders gives

$$(1+\eta)^{-1/2} = 1 + \frac{\hat{R}\cdot\vec{r}_i}{R} + \left[\frac{3}{2}\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\frac{r_i^2}{R^2}\right] + \mathcal{O}\left(\frac{r_i^3}{R^3}\right).$$

Multipole expansion of the potential. Substituting this series into the potential gives, to second order in r_i/R,

$$\boxed{\phi(\vec{R}) \approx \sum_{i=1}^{N} \frac{q_i}{R} \left\{ 1 + \frac{\hat{R}\cdot\vec{r}_i}{R} + \left[\frac{3}{2}\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\frac{r_i^2}{R^2} \right] \right\}.} \tag{1.1}$$

The three terms in braces are the monopole, dipole, and quadrupole terms.

Monopole term: the total charge. The first term is the monopole term,

$$\boxed{\sum_{i=1}^{N} \frac{q_i}{R} = \frac{Q}{R}, \qquad Q = \sum_{i=1}^{N} q_i,} \tag{1.2}$$

where Q is the total charge, also called the monopole moment, a scalar or rank-0 tensor. This term vanishes when the point cloud is neutral.

Dipole term: the dipole moment vector. The second term is the dipole term,

$$\boxed{\sum_{i=1}^{N} \frac{q_i\,(\hat{R}\cdot\vec{r}_i)}{R^2} = \frac{\hat{R}\cdot\vec{p}}{R^2}, \qquad \vec{p} = \sum_{i=1}^{N} q_i\,\vec{r}_i,} \tag{1.3}$$

where $\vec{p}$ is the dipole, a vector or rank-1 tensor. It is usually called the dipole moment vector. This term remains even when the point cloud is neutral.

Quadrupole term written as a matrix product. The third term is the quadrupole term,

$$\sum_{i=1}^{N} \frac{q_i}{R} \left[\frac{3}{2}\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\frac{r_i^2}{R^2} \right] = \frac{1}{2R^3} \sum_{i=1}^{N} q_i \left[3(\hat{R}\cdot\vec{r}_i)^2 - r_i^2 \right].$$

From now on, Greek indices $\alpha, \beta \in \{x, y, z\}$ label Cartesian components, and the Latin index i still labels the charges. For example, $r_{i\alpha}$ is the α component of $\vec{r}_i$. In components, the squared dot product becomes a double sum,

$$(\hat{R}\cdot\vec{r}_i)^2 = \sum_{\alpha} \hat{R}_\alpha r_{i\alpha} \sum_{\beta} \hat{R}_\beta r_{i\beta} = \sum_{\alpha,\beta} \hat{R}_\alpha \hat{R}_\beta\, r_{i\alpha} r_{i\beta}.$$

Because $\hat{R}$ is a unit vector, $\sum_{\alpha,\beta} \hat{R}_\alpha \hat{R}_\beta \, \delta_{\alpha\beta} = 1$, where $\delta_{\alpha\beta}$ is the Kronecker delta. The squared distance therefore takes the same form,

$$r_i^2 = \sum_{\alpha,\beta} \hat{R}_\alpha \hat{R}_\beta \, r_i^2 \, \delta_{\alpha\beta}.$$

Both sums resemble the inner product $\vec{u}^T M \vec{u}$ of a vector $\vec{u}$ and a matrix M. To see this, let $\vec{w} = M\vec{u}$. Each component $w_\alpha = \sum_\beta M_{\alpha\beta} u_\beta$ multiplies the α-th row of M entry by entry with $\vec{u}$ and sums over the columns β. The inner product is then

$$\vec{u}^T M \vec{u} = \vec{u}^T \vec{w} = \sum_\alpha u_\alpha w_\alpha = \sum_{\alpha,\beta} u_\alpha M_{\alpha\beta} u_\beta.$$

Quadrupole moment tensor. Combining the two sums with $u_\alpha = \hat{R}_\alpha$ gives

$$\sum_{i=1}^N q_i \left[3(\hat{R} \cdot \vec{r}_i)^2 - r_i^2 \right] = \sum_{\alpha,\beta} \hat{R}_\alpha \hat{R}_\beta \sum_{i=1}^N q_i \left(3 \, r_{i\alpha} r_{i\beta} - r_i^2 \, \delta_{\alpha\beta} \right) = \hat{R}^T \mathcal{Q} \, \hat{R}.$$

The quadrupole term is therefore

$$\boxed{\begin{gathered} \sum_{i=1}^N \frac{q_i}{R} \left[\frac{3}{2} \frac{(\hat{R} \cdot \vec{r}_i)^2}{R^2} - \frac{1}{2} \frac{r_i^2}{R^2} \right] = \frac{\hat{R}^T \mathcal{Q} \, \hat{R}}{2R^3}, \\ \mathcal{Q}_{\alpha\beta} = \sum_{i=1}^N q_i \left(3 \, r_{i\alpha} r_{i\beta} - r_i^2 \, \delta_{\alpha\beta} \right), \end{gathered}} \tag{1.4}$$

where $\mathcal{Q}$ is the quadrupole, a symmetric matrix or rank-2 tensor. It is usually called the quadrupole moment tensor. Its trace vanishes, because $\sum_\alpha \left(3 \, r_{i\alpha}^2 - r_i^2 \right) = 0$ for every charge.

Dependence of the moments on the choice of origin. The multipole moments depend on where the origin is placed. Shifting the origin by $\vec{d}$ replaces $\vec{r}_i$ with $\vec{r}_i - \vec{d}$, so the dipole moment becomes

$$\vec{p}\,' = \sum_{i=1}^N q_i \left(\vec{r}_i - \vec{d} \right) = \vec{p} - Q \, \vec{d}.$$

The dipole moment is therefore independent of the origin only when $Q = 0$. The same argument shows that $\mathcal{Q}$ is independent of the origin only when Q and $\vec{p}$ both vanish. In general, only the lowest nonvanishing moment is independent of the origin.

2 Polarization

Why the dipole moment describes a neutral material in a field. Most materials are electrically neutral. Under an external electric field, the total charge remains zero, but the electrons and the nuclei are displaced in opposite directions and produce an induced dipole. Some materials, such as polar molecular liquids and ferroelectrics, also carry permanent dipoles even without an external field. In both cases, the monopole term vanishes, so the dipole term is the leading term of the multipole expansion far from the material. Because $Q = 0$, the dipole moment is also independent of the choice of origin, as shown in the previous section. The dipole moment is therefore the quantity of interest whenever a material interacts with an electric field.

Polarization as a dipole moment density, and why it is not a derivative. The polarization vector $\vec{P}$ describes the dipole moment density of a material. Real materials consist of discrete dipoles, so the common definition $\vec{P} = \Delta\vec{p}/\Delta V$ cannot be a derivative: the ratio does not converge as the volume $\Delta V \to 0$. A small volume between atoms contains no dipole and gives zero, whereas a small volume around a point dipole gives a ratio that grows without bound. The ratio is meaningful only for a volume that is large compared with the interatomic distance a but small compared with the length L over which the macroscopic fields vary.

Polarization as a spatial average of the microscopic dipoles. The precise definition replaces the limit with an average. A set of point dipoles $\vec{p}_k$ at positions $\vec{r}_k$ has the microscopic dipole density

$$\vec{P}_{\text{micro}}(\vec{r}) = \sum_k \vec{p}_k \, \delta(\vec{r} - \vec{r}_k),$$

which is a distribution rather than a smooth function. Averaging it with a smooth weighting function w, normalized as $\int w(\vec{r})\, d^3r = 1$ and with a width ℓ that satisfies $a \ll \ell \ll L$, gives the macroscopic polarization

$$\boxed{\vec{P}(\vec{r}) = \int w(\vec{r} - \vec{r}\,')\, \vec{P}_{\text{micro}}(\vec{r}\,')\, d^3r' = \sum_k \vec{p}_k \, w(\vec{r} - \vec{r}_k).} \tag{2.1}$$

This $\vec{P}$ is a density in the integral sense. Over a macroscopic region Ω, the integral $\int_\Omega w(\vec{r} - \vec{r}_k)\, d^3r$ is close to one for a dipole deep inside Ω and close to zero for a dipole far outside it. Only dipoles within a distance ℓ of the boundary contribute partially. The dipole moment of the region is therefore

$$\boxed{\vec{p}_\Omega = \int_\Omega \vec{P}(\vec{r})\, d^3r.} \tag{2.2}$$

Figure 1 illustrates this averaging for a one-dimensional chain of dipoles.

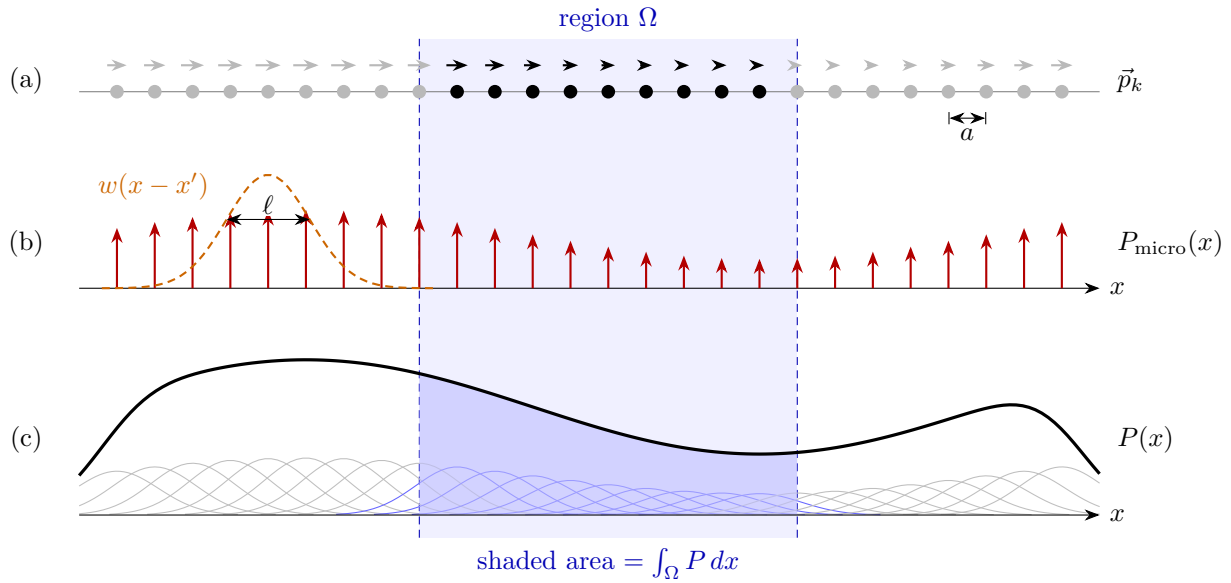


Figure 1: Averaging of a one-dimensional chain of dipoles. (a) Discrete dipoles $\vec{p}_k$ with interatomic distance a; dipoles inside Ω in black. (b) Microscopic dipole density as delta peaks, with one averaging window w of width $\ell \gg a$. (c) Contributions $p_k\, w(x - x_k)$ of single dipoles (thin curves) and their sum, the smooth polarization $P(x)$ (thick curve). Dipoles far outside Ω (gray) do not contribute to the shaded area, and dipoles within ℓ of the boundary contribute partially.

Limitation of the averaging picture in periodic crystals. The averaging assumes that the charge can be grouped into localized dipoles $\vec{p}_k$. In a periodic crystal, the electron density is continuous, so the dipole moment of a unit cell depends on how the cell is chosen. Different choices give bulk polarizations that differ by integer multiples of a polarization quantum, which is proportional to a lattice vector divided by the cell volume. Only changes in $\vec{P}$, for example under strain or atomic displacement, are physical. The modern theory of polarization computes these changes from the Berry phase of the Bloch wave functions (King-Smith and Vanderbilt, 1993; Resta, 1994).

Electric field of a single dipole. The field of a volume of dipoles is built from the field of a single dipole. For a point dipole $\vec{p}$ at the origin, the dipole term of the multipole expansion gives the potential

$$\phi(\vec{R}) = \frac{\hat{R}\cdot\vec{p}}{R^2} = \frac{\vec{p}\cdot\vec{R}}{R^3}.$$

The electric field is the negative gradient of the potential, $\vec{E} = -\vec{\nabla}\phi$. With $\vec{\nabla}(\vec{p}\cdot\vec{R}) = \vec{p}$ and $\vec{\nabla}R^{-3} = -3\vec{R}/R^5$,

$$\vec{E}(\vec{R}) = -\frac{\vec{p}}{R^3} + \frac{3(\vec{p}\cdot\vec{R})\,\vec{R}}{R^5}.$$

The field of a point dipole is therefore

$$\boxed{\vec{E}(\vec{R}) = \frac{3(\vec{p}\cdot\hat{R})\,\hat{R} - \vec{p}}{R^3}.} \tag{2.3}$$

Potential of a volume of dipoles. Because $\vec{P}$ is a smooth density, the volume element d^3r at the source point $\vec{r}$ carries the dipole moment $\vec{P}(\vec{r})\,d^3r$. A dipole at $\vec{r}$ instead of the origin has the potential $\vec{p}\cdot(\vec{R}-\vec{r})/|\vec{R}-\vec{r}|^3$. Summing the contributions of all volume elements of a material occupying Ω gives

$$\phi(\vec{R}) = \int_\Omega \frac{\vec{P}(\vec{r})\cdot(\vec{R}-\vec{r})}{|\vec{R}-\vec{r}|^3}\,d^3r.$$

From now on, $\vec{\nabla}_{\vec{r}}$ denotes the gradient with respect to the source point $\vec{r}$, whereas $\vec{\nabla}$ without a subscript still acts on the field point $\vec{R}$. The integrand simplifies with the identity

$$\vec{\nabla}_{\vec{r}}\frac{1}{|\vec{R}-\vec{r}|} = \frac{\vec{R}-\vec{r}}{|\vec{R}-\vec{r}|^3},$$

so that

$$\phi(\vec{R}) = \int_\Omega \vec{P}\cdot\vec{\nabla}_{\vec{r}}\frac{1}{|\vec{R}-\vec{r}|}\,d^3r.$$

Bound charges as the sources of the polarization potential. The product rule $\vec{\nabla}_{\vec{r}}\cdot(f\vec{P}) = f\,\vec{\nabla}_{\vec{r}}\cdot\vec{P} + \vec{P}\cdot\vec{\nabla}_{\vec{r}}f$ with $f = 1/|\vec{R}-\vec{r}|$ splits the integrand into two parts,

$$\phi(\vec{R}) = \int_\Omega \vec{\nabla}_{\vec{r}}\cdot\left(\frac{\vec{P}}{|\vec{R}-\vec{r}|}\right)d^3r - \int_\Omega \frac{\vec{\nabla}_{\vec{r}}\cdot\vec{P}}{|\vec{R}-\vec{r}|}\,d^3r.$$

The divergence theorem turns the first integral into an integral over the boundary $\partial\Omega$, with outward unit normal $\hat{n}$ and surface element dS. The potential of the polarized material is therefore

$$\boxed{\begin{gathered}\phi(\vec{R}) = \oint_{\partial\Omega}\frac{\sigma_b(\vec{r})}{|\vec{R}-\vec{r}|}\,dS + \int_\Omega \frac{\rho_b(\vec{r})}{|\vec{R}-\vec{r}|}\,d^3r, \\ \sigma_b = \vec{P}\cdot\hat{n}, \qquad \rho_b = -\vec{\nabla}_{\vec{r}}\cdot\vec{P},\end{gathered}} \tag{2.4}$$

where σ_b is the bound surface charge density and ρ_b is the bound volume charge density. A polarized material thus produces the same potential as these two charge distributions. In particular, a uniform polarization has $\rho_b = 0$ and produces only a surface charge. Figure 2 shows the origin of both bound charges in a chain of dipoles.

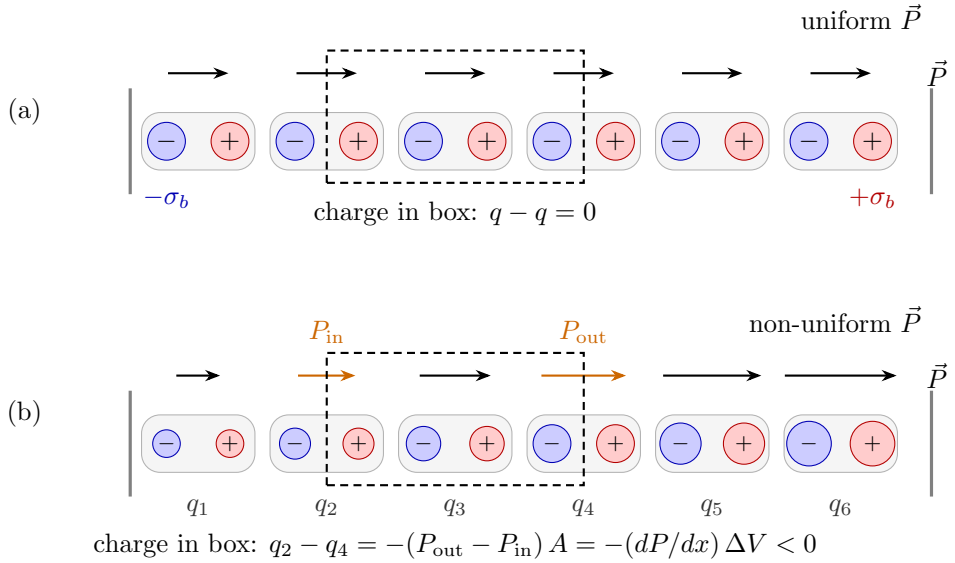


Figure 2: Bound charges of a chain of dipoles between two surfaces. (a) Uniform polarization: the charges of neighboring dipoles cancel inside the box, and only the surfaces carry the uncompensated charges $\pm\sigma_b$. (b) Polarization increasing to the right: the box holds $+q_2$ and $-q_4$ with $q_4 > q_2$, so it carries the net charge $-(dP/dx)\,\Delta V$. Here A is the cross-sectional area and ΔV the volume of the box.

3 Dielectric Sphere inside a Dielectric Medium

3.1 Gauss's Law in a Polarizable Material

Electric displacement as the field of the free charges. Gauss's law counts both the free charge ρ_f and the bound charge ρ_b,

$$\vec{\nabla} \cdot \vec{E} = 4\pi(\rho_f + \rho_b), \qquad \rho_b = -\vec{\nabla} \cdot \vec{P}.$$

The bound charge is unknown until $\vec{E}$ is known. Moving it to the left side defines the electric displacement $\vec{D}$,

$$\boxed{\vec{D} = \vec{E} + 4\pi\vec{P}, \qquad \vec{\nabla} \cdot \vec{D} = 4\pi\rho_f.} \tag{3.1}$$

This is only a change of variables; a relation between $\vec{P}$ and $\vec{E}$ is still needed.

Linear response and the dielectric tensor. For a weak field, the polarization is expanded to first order about $\vec{E} = 0$,

$$P_\alpha(\vec{E}) = P_\alpha(0) + \sum_\beta \chi_{\alpha\beta} E_\beta + \mathcal{O}(E^2), \qquad \chi_{\alpha\beta} = \left.\frac{\partial P_\alpha}{\partial E_\beta}\right|_{\vec{E}=0}.$$

With $P_\alpha(0) = 0$ and without the $\mathcal{O}(E^2)$ terms,

$$\boxed{P_\alpha = \sum_\beta \chi_{\alpha\beta} E_\beta, \qquad D_\alpha = \sum_\beta \varepsilon_{\alpha\beta} E_\beta, \qquad \varepsilon_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi\chi_{\alpha\beta},} \tag{3.2}$$

where $\chi_{\alpha\beta}$ is the susceptibility tensor and $\varepsilon_{\alpha\beta}$ is the dielectric tensor. Here $\vec{E}$ is the macroscopic field at the same point, not the external field.

Isotropic materials and the dielectric constant. An isotropic material responds equally in every direction, $\chi_{\alpha\beta} = \chi\,\delta_{\alpha\beta}$, so that

$$\boxed{\vec{P} = \chi\,\vec{E}, \qquad \vec{D} = \varepsilon\,\vec{E}, \qquad \varepsilon = 1 + 4\pi\chi,} \tag{3.3}$$

where χ is the electric susceptibility and ε is the dielectric constant.

Limits of the linear model. Each assumption above fails in some materials. A ferroelectric carries a spontaneous polarization, $\vec{P}(0) = \vec{P}_s \neq 0$. A strong field, as in nonlinear optics, makes the second-order term significant,

$$P_\alpha = \sum_\beta \chi_{\alpha\beta} E_\beta + \sum_{\beta,\gamma} \chi^{(2)}_{\alpha\beta\gamma} E_\beta E_\gamma + \mathcal{O}(E^3).$$

A time-dependent field gives a frequency-dependent response, $\vec{P}(\omega) = \chi(\omega)\,\vec{E}(\omega)$; for example, metals at optical frequencies have $\varepsilon(\omega) < 0$. This note treats static, weak fields in linear isotropic media only.

3.2 Boundary Conditions between Two Dielectric Media

Laplace's equation inside each medium. Consider two linear isotropic media, labeled 1 and 2, with constant dielectric constants ε_1 and ε_2 and $\rho_f = 0$. With $\vec{D} = -\varepsilon\vec{\nabla}\phi$,

$$\vec{\nabla}\cdot\vec{D} = -\varepsilon\,\nabla^2\phi = 0,$$

so the potential satisfies Laplace's equation in both media,

$$\boxed{\nabla^2\phi_1 = 0, \qquad \nabla^2\phi_2 = 0.} \tag{3.4}$$

Normal component from the divergence theorem. Let $\hat{n}$ be the unit normal to the interface, pointing from medium 1 into medium 2, with $\partial/\partial n = \hat{n}\cdot\vec{\nabla}$ and $D_n = \vec{D}\cdot\hat{n}$. A pillbox of face area ΔA and height h straddles the interface. The divergence theorem gives

$$\oint_{\text{pillbox}} \vec{D}\cdot d\vec{A} = \int_{\text{pillbox}} \vec{\nabla}\cdot\vec{D}\; d^3r = 4\pi Q_f = 0.$$

As $h \to 0$, the side wall drops out and only the two faces remain,

$$(D_{2,n} - D_{1,n})\,\Delta A = 0 \quad \Rightarrow \quad \varepsilon_1 E_{1,n} = \varepsilon_2 E_{2,n}.$$

With $E_n = -\partial\phi/\partial n$,

$$\boxed{\varepsilon_1 \frac{\partial\phi_1}{\partial n} = \varepsilon_2 \frac{\partial\phi_2}{\partial n}.} \tag{3.5}$$

The same pillbox applied to $\vec{E}$ encloses the bound surface charge, $E_{2,n} - E_{1,n} = 4\pi\sigma_b$, so E_n itself jumps across the interface.

Tangential component from Stokes' theorem. In electrostatics, $\partial\vec{B}/\partial t = 0$, so Faraday's law gives $\vec{\nabla}\times\vec{E} = 0$. Let C be a rectangular loop with two sides of length Δl along a tangent $\hat{t}$ of the interface, one in each medium, and two sides of height h crossing it. Stokes' theorem gives

$$\oint_C \vec{E}\cdot d\vec{l} = \int_S \left(\vec{\nabla}\times\vec{E}\right)\cdot d\vec{S} = 0,$$

where S is the surface bounded by C. As $h \to 0$, only the two long sides remain,

$$(E_{2,t} - E_{1,t})\,\Delta l = 0 \quad \Rightarrow \quad E_{1,t} = E_{2,t}.$$

The same theorem forbids closed field lines, because $\vec{E}\cdot d\vec{l} > 0$ along a field line and $\oint \vec{E}\cdot d\vec{l}$ would be positive. Across the interface, the potential difference over the height h vanishes for a finite field,

$$\phi_2 - \phi_1 = -\int_0^h \vec{E}\cdot\hat{n}\; dl \;\longrightarrow\; 0 \qquad (h \to 0),$$

which gives

$$\boxed{\phi_1 = \phi_2 \quad \text{on the interface.}} \tag{3.6}$$

Since $E_t = -\hat{t}\cdot\vec{\nabla}\phi$ and $\phi_1 = \phi_2$ holds all along the interface, this condition contains $E_{1,t} = E_{2,t}$.